Test Task Report

Project : Equivariant Vision Networks for Predicting Planetary Systems' Architectures

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Repository:	https://github.com/ShreyPatel1311/EXXA-GSoC-2026
Models:	https://huggingface.co/Godseye1311/EXXA-MAE-Models

1. Data Preprocessing and Augmentation

Dataset: Synthetic continuum observations of ALMA(1250 microns)[1] (Terry et al., 2022) – 100 images of protoplanetary disks with no labels are provided. This data, generated by Terry et al., was created using particle hydrodynamics simulations using PHANTOM SPH code.

Preprocessing: Every image in the dataset have been processed through power law transformation, which will enhance the features in the image.

Data Augmentation Strategy: After Image Enhancement, the images in the dataset are randomly rotated, flipped, cropped and resized, each variant is stored in the dataset.

2. General Test – Clustering

Approach: Figure 1, shows the pipeline for creating clusters using Masked Auto-Encoder, UMAP and HDBSCAN Algorithms.

- Masked Auto-Encoder(or MAE) will extract the features from image to latent space. Charles Fox, shows that MAE performs better on non-labeled data for feature extraction by only using 25% of patches [2].
- After Latent space extracts features, these features are passed on to UMAP lock for dimensionality reduction.
- These vectors after dimension reduction are passed on to HDBSCAN algorithm for clustering. HDBSCAN performs better than most clustering algorithm[3] (Mclennes 2017).
- MAE will have a Cross-Attention Decoder, which will try to reconstruct the input image from latent space from Encoder.

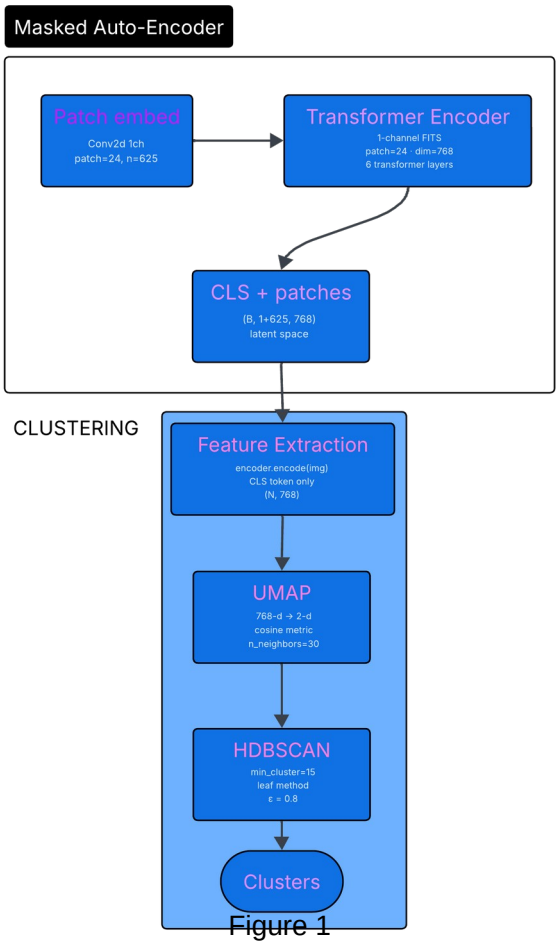


Figure 1

3. Image Based Test

Approach: Middle image in Figure 2, shows the 2 paths for image reconstruction, one uses MAE Decoder and other uses Cross-Attention Decoder as discussed above in last section.

- In this test, 2 different decoders are used and their performances are compared. They are evaluated on the basis of how good image they are able to reconstruct and how well clusters are made, keeping the MAE Encoder same.

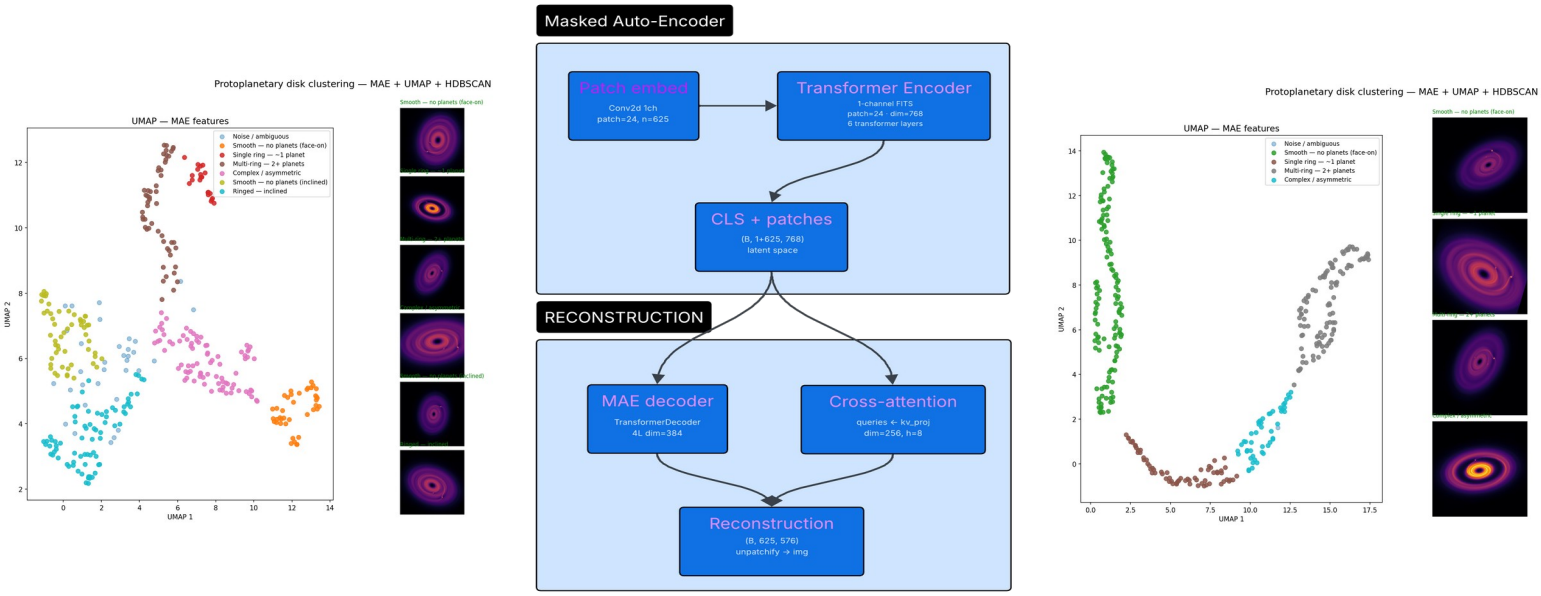
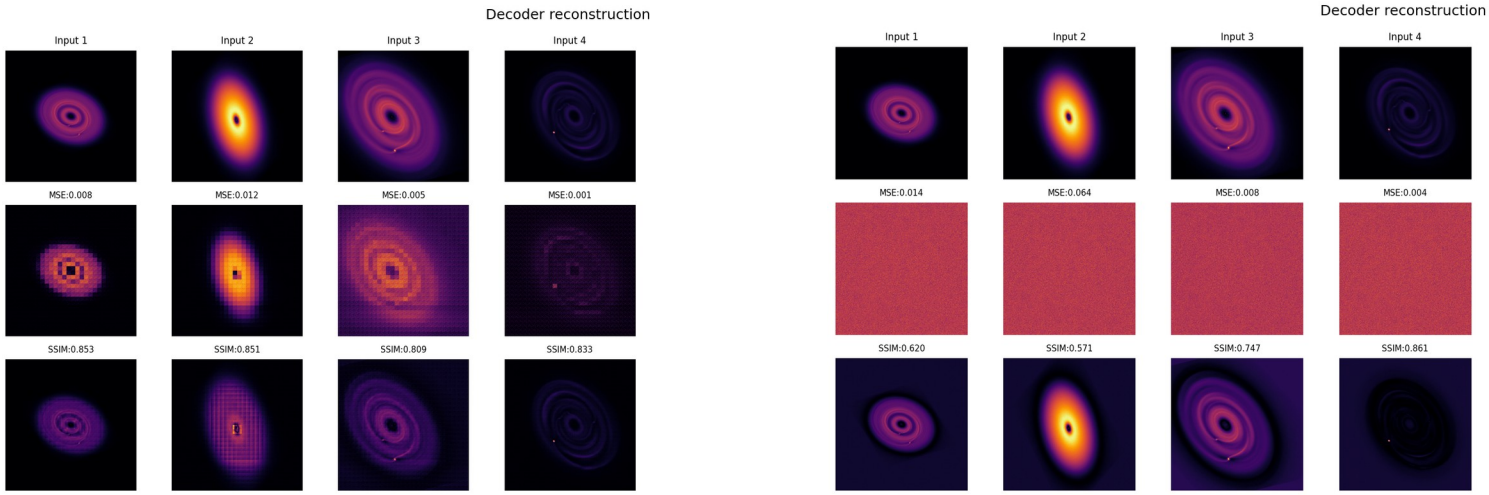


Figure 2. *Center: Image Reconstruction Pipeline. Left: Clustering due to Cross Attention Decoder. Right: Clustering due to MAE Decoder.*



4. Results

- Results are evaluated on the basis of clusters and images reconstructed. Image Reconstruction can be measured using Mean Squared Error and Multi-Scale Structural Similarity Index Measure. Images shown in Figure 3 are the images reconstructed by Masked Auto-Encoder. On the Left,

the images are reconstructed using MAE Decoder whereas, on the right, they are developed using Cross-Attention Decoder. Similarly, the clusters in Figure 2 were made by the respective decoders only.

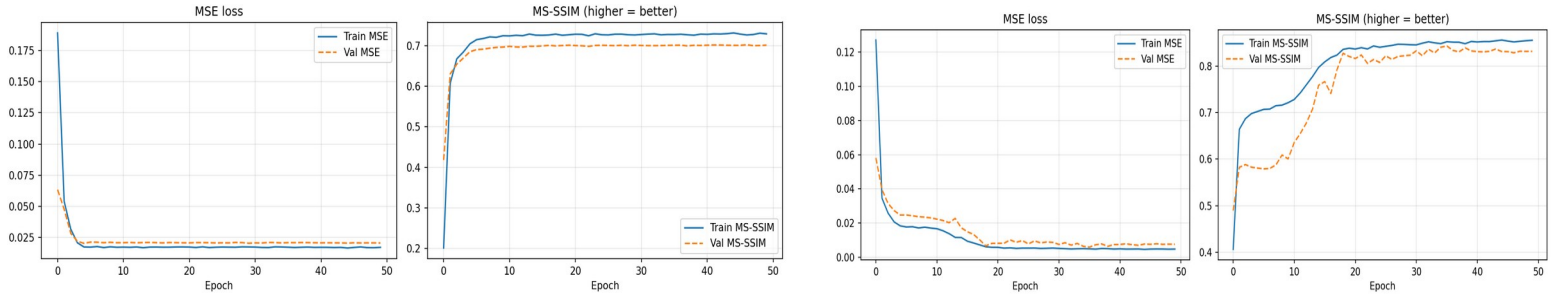


Figure 4. *The training logs under MSE and MS-SSIM respectively. Left(2):Cross-Attention Decoder, Right(2): MAE Decoder*

4. Key Findings

Image Reconstruction by MAE Decoder. As shown in Figure 3 (left), MAE Decoder creates blurry images whereas Cross-Attention constructs noisy image. When looked carefully, the latter produces poor image even under ssim.

Clustering. For clustering purposes, MAE Encoder trained with cross attention as decoder groups these images well as compared to MAE Decoder.

Power Law Preprocessing and Data Augmentation. Enhanced the hidden features in darker images of disks, which in the end helped model to converge early in training. Although, augmented data performed better, more aggressive data augmentation techniques or real data are required for better image reconstruction.

5. Proposed Improvements for the GSoC Project

- **Aggressive Data Preprocessing.** More data can be augmented by techniques like random brightness and contrast shift of image or assigning calculated values in power law transformation. Center Cropping must be implemented, as clusters made mistake stars as planets.
- **Loss Function Exploration.** MAE Decoder can use a mixture of MSE, MS-SSIM and L1 loss for smoother outputs for image reconstruction, or any other variants of MSE.
- **Different clustering tecniques.** Other clustering techniques can also be implemented such as spectral clustering or Deep Embeded Clustering(DEC).
- **State Space Models(SSM).** Variant of Mamba architecture can be explored for feature extraction or image reconstruction.

References

[1] J. P. Terry *et al*. Locating Hidden Exoplanets in ALMA Data Using Machine Learning. DOI 10.3847/1538-4357/aca477

[2] Charles Fox, AstroMAE: Redshift Prediction Using a Masked Autoencoder with a Novel Fine-Tuning Architecture arXiv:2409.01825v1

[3] McInnes, Leland & Healy, John & Astels, Steve. (2017). hdbscan: Hierarchical density based clustering. The Journal of Open Source Software. 2. 205. 10.21105/joss.00205.